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Paleomagnetism of the Mid-Tertiary Ayacara Formation, southern Chile: Counterclockwise rotation in a dextral shear zone

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Abstract—Relative motions of the southern Chilean forearc are assessed through a paleomagnetic study of the Eocene-Miocene Ayacara Formation. This marine volcaniclastic unit was studied at three localities, all immediately west of the Liquiñe-Ofqui fault zone (LOFZ), a 1000-km-long, NS-trending intracontinental shear zone with probable Cenozoic right-lateral displacement. A paleomagnetic pole (74.8S; 44.0E; A-95, 8.5°) was calculated by combining 11 site-mean VGP from the Ayacara Fm. with three sites from the nearby Oligocene Cocotue Beach basalts (previously published). When compared with a 20-40 Ma North American cratonic pole rotated into the South American reference framework this pole implies counterclockwise rotation of 14.6±11.8 . We speculate that this counterclockwise rotation results from processes that tend to impel a detached sliver of continental crust northward along the continental margin, against a buttress. In our model, northward displacement of the trailing edge of the sliver is accomplished by segmenting the sliver into lens-shaped blocks which rotate counterclockwise as they move northward. The geometry of the LOFZ clearly supports this model, which may have application to other buttressed strike-slip fault zones.

Resumen—Los movimientos relativos del antearco de Chile meridional son evaluados mediante un estudio paleomagnético de la Formación Ayacara (Eoceno-Mioceno). Esta unidad volcanoclástica marina fue estudiada en tres localidades, todas ellas ubicadas inmediatamente al oeste de la Zona de Falla Liquiñe-Ofqui (ZFLO): una zona de cizalle intracontinental de rumbo NS, con probable movimiento dextral cenozoico. Un polo paleomagnético (74.8S; 44.0E; A-95, 8.5°) fue calculado a partir de la combinación de 11 PGV de la Formación Ayacara y de tres sitios provenientes de los basaltos oligocenos de Playa Cocotué (previamente publicado). Al compararlo con un polo del cratón norteamericano (20-40 Ma), rotado al sistema de referencia sudaméricano, se obtiene una rotación antihoraria de 14.6±11.8. Se especula que esta rotación antihoraria resulta de procesos que tienden a empujar un panel cortical despegado hacia el norte, a lo largo del margen continental, contra un obstáculo. En nuestro modelo , el desplazamiento hacia el norte se logra mediante la segmentación del panel en bloques lenticulares que rotan de manera antihoraria al moverse hacia el norte. La geometría de la sección meridional de la ZFLO apoya claramente este modelo, el cual puede aplicarse a otras zonas de cizalle de rumbo obstaculizadas.

INTRODUCTION

PALEOMAGNETIC DATA provide evidence for many hundreds of kilometers of coastwise transport of continental slivers along the western North American Cordillera during the Mesozoic and early Cenozoic (e.g., Beck, 1980, 1989; Irving *et al.*, 1985). Oblique subduction has been proposed as a driving mechanism for such tectonic transport (Fitch, 1972; Beck, 1986; Jarrard, 1986) and is consistent with plate reconstructions for the northeastern Pacific (Engebretson *et al.*, 1985). Similar oblique subduction of the Farallon (Nazca) plate beneath the South American plate during the last 60 m.y. (Cande and Leslie, 1986; Pardo-Casas and Molnar, 1987), and the existence of major trench-parallel lineaments along the Chilean Andes, suggest that north-south translation of outboard pieces of the Andean forearc also might have occurred. However, paleomagnetic studies to date have not reliably detected any such displaced blocks. The Chilean forearc south of about 42° S contains several outcrops of Eocene-Miocene marine volcaniclastic rocks, grouped into the Ayacara Formation by Levi *et al.* (1966). These all lie immediately west of the Liquiñe Ofqui fault zone (LOFZ), a 1000-km-long crustal lineament that cuts the main Andean arc and forearc between about 39° and 47° S (Hervé and Thiele, 1987). The history of the LOFZ is not known in detail, and several current studies are attempting to unravel its tectonic evolution. Of great interest is the amount of lateral displacement and how that displacement is accommodated.

REGIONAL GEOLOGY

The Andes between 42° and 43° S are characterized by roughly defined north-south belts of igneous, metamorphic and sedimentary rocks (Fig. 1).

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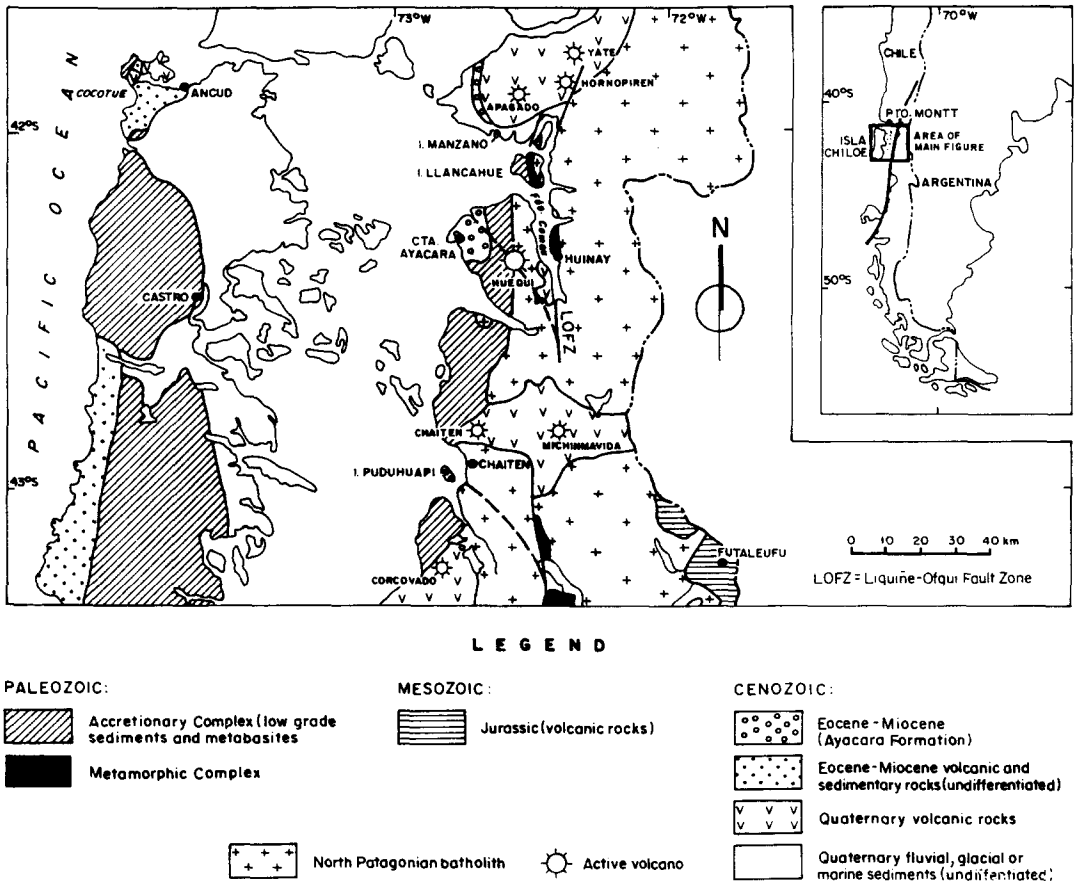


Figure 1. Regional Geological map from the Chilean Andes between 41°30´ and 42°30´S. Modified after Pankhurst *et al.* (1992) and Valenzuela (1982).

The westernmost belt, Chiloé Island, is a prolongation of the Chilean Coast Range. It consists mainly of low grade metasedimentary rocks interpreted as a Paleozoic forearc accretionary complex (Godoy *et al.*, 1984; Hervé, 1988). Locally this accretionary complex is overlain unconformably by Eocene-Miocene marine sedimentary and felsic volcanic deposits (García, 1968; Valenzuela, 1982; García *et al.*, 1988).

To the east, the Main Range consists mostly of plutonic rocks of the North Patagonian batholith (NPB). The western portion of the NPB is dominated by Mio-Pliocene granodiorite and tonalite; to the east it consists mainly of monzogranite and monzonite that yield mid-Cretaceous radiometric ages (Pankhurst *et al.*, 1992). Locally the western batholithic rocks intrude the Eocene-Miocene layered rocks (Ayacara Fm.), although in other localities they are in fault contact (Cembrano, 1990). Locally Plio-Pleistocene volcanic deposits rest on all the lithological units mentioned earlier.

The Liquiñe-Ofqui Fault Zone (LOFZ)

The LOFZ has been regarded by most researchers as a dextral strike-slip fault (e.g., Hervé *et al.*, 1976; Forsythe and Nelson, 1985; Beck, 1988), although in many places (particularly where it runs through plutonic rocks of the North Patagonian batholith) sense and amount of displacement are difficult to assess. A general description of the LOFZ is given by Hervé and Thiele (1987). On the basis of distinctive structural and geologic features they have divided the LOFZ into the following segments:

From 39°30' to 41°30' the LOFZ is represented by one main lineament striking N10E (Fig. 2). At Liquiñe, it is described by Hervé (1976) as a 3-km-wide belt of mylonitic and cataclastic rocks. Right-lateral motion along the LOFZ has been inferred from mesoscopic faults within the mylonitic zone. A poorly constrained age of 28 Ma was obtained in a dacite porphyry which crosscuts the LOFZ-related mylonitization (Hervé, 1976).

From about 41°30' to 42° the LOFZ shows more sporadic and narrower cataclastic strips than further north. In this zone recent geochronology (Drake *et al.*, 1990) shows that the LOFZ juxtaposes Miocene and Lower Cretaceous intrusive rocks.

Between 42° to 43°, the westernmost portion of the NPB and an associated 2-3 km wide aureole of medium to high grade metamorphic rocks are located within and east of the LOFZ. The plutonic rocks, which are as young as Pliocene (Pankhurst *et al.*, 1992), show field relations and microstructural features consistent with brittle-ductile deformation in a dextral shear zone (Cembrano, 1990; 1992). These relatively deep crustal levels are exposed immediately east of Eocene-Miocene volcaniclastic rocks and may indicate post-Pliocene fault-controlled differential uplift.

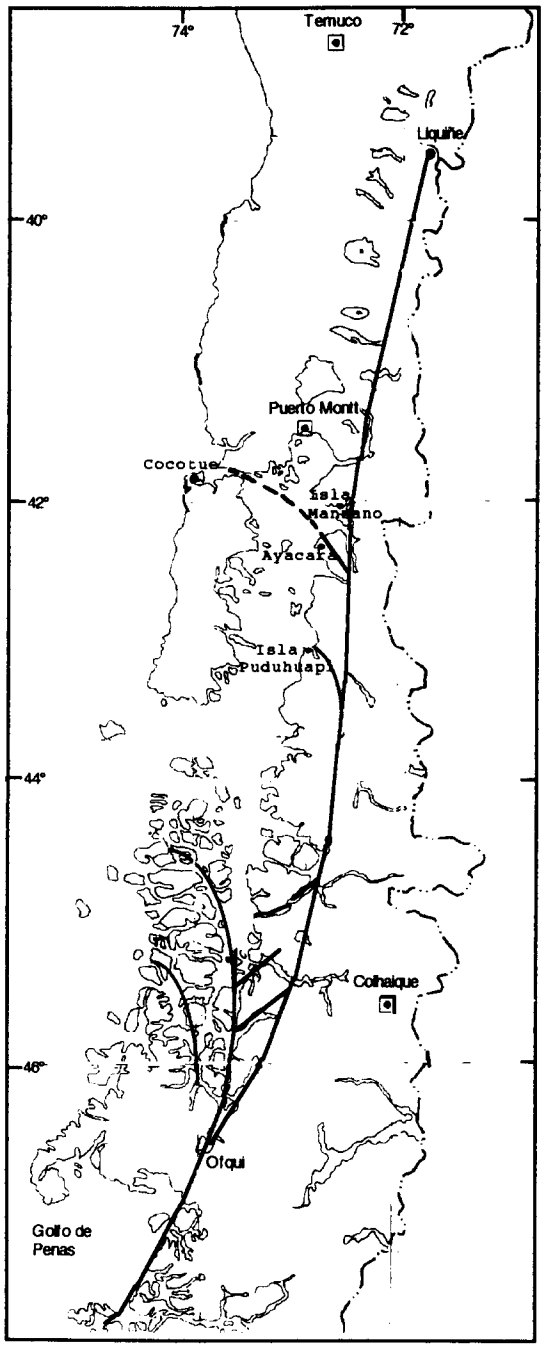


Figure 2. Main lineaments of the Liquiñe-Ofqui fault zone. Modified from Hervé and Thiele, 1987.

From 43° S southward the LOFZ consists of a single linear feature that gives way to a series of en echelon NW and NE-trending lineaments, that (together with the main NNE-trending LOFZ) outline a series of curved fault segments that are concave toward the west. Isolated, relatively undeformed sequences of Paleogene (?) volcanosedimentary rocks (Estratos de La Junta) and Pleistocene lacustrine sediments (Estratos de La Silla) in places fill the glacial valley carved on the LOFZ south of 43°00' S. A 1.2 Ma basaltic flow (Hervé *et al.*, 1990) lies on top of this sedimentary fill at 43°30' S, overlapping the LOFZ's main trace.

At its southernmost limit, the LOFZ appears to originate in the Golfo de Penas, which has been considered to be an extensional (trailing edge) Cenozoic basin (Forsythe and Nelson,1985).

The tectonic significance of the LOFZ is hotly debated. Hervé (1976), Leslie (1986) and Beck (1988) propose that oblique subduction of the Nazca plate beneath South America caused dextral shear in the leading edge of the overriding plate, leading to formation of the LOFZ; the model invoked is similar to Fitch's (1972) explanation for the Sumatra fault. Leslie (1986) identified a period of highly oblique subduction during the Paleogene and suggested that the activity of the LOFZ might have reached a peak at that time. After a period of almost orthogonal convergence between 26 and 20 Ma, subduction of the Nazca plate has been slighly right-oblique (Pardo Casas and Molnar, 1987). This scenario may have favored dextral transpression along the LOFZ until recent times.

Forsythe and Nelson (1985), on the other hand, proposed that the LOFZ originated when the Chile Rise, acting as a rigid indenter, impinged on the South American plate. Their model predicts considerable Neogene offset on the fault.

The Ayacara Formation

Levi *et al.* (1966) described the Ayacara Formation as "a well consolidated volcaniclastic sequence cropping out in Caleta Ayacara" (Fig. 1). They identified two members: a lower member, 200 m thick, consisting of conglomerate and sandstone, and an upper member, 350 m thick, of tuff, dacitic volcanic sandstone and shale with synsedimentary structures suggesting turbidity current deposits. The same authors described a 150-m-thick sedimentary sequence, also assigned by them to the Ayacara Formation, that crops out on Isla Manzano (Fig. 1). Microfossils found at this locality suggest an Eocene-Miocene age (Martinez, 1961). Both the Caleta Ayacara and Isla Manzano exposures are gently to steeply dipping, nearly homoclinal, and show little evidence of internal deformation (cross-faults, etc.). There is no evidence of hydrothermal alteration, deep weathering, re-heating, or other processes that might lead to remagnetization.

A third outcrop area, located on Isla Puduhuapi (Fig. 1), consists of a subhorizontal sequence of shale and andesite. This section was informally named the "Puduhuapi strata" by Levi *et al.* (1966) and was assigned a general Tertiary age on the basis of lithologic similarities with strata from the Ayacara Formation.

Individual sections from the three Ayacara localities have never been formally correlated, although preliminary attempts to do so were made by Levi *et al.* (1966), Solano (1978), Araya (1979), and Fuenzalida (1979). During the present study, the lithology, age, and thickness of the Formation at its type locality and at Isla Manzano were compared (Fig. 3).

The section from Caleta Ayacara shows a 100-m-thick lower member of conglomerate and tuffaceous sandstone. The upper member consists of 500 m of tuff, tuffaceous sandstone and tuffaceous siltstone. A facies analysis indicates that the lower member may have been deposited by debris flows. The upper member includes pyroclastic tur-

bidity current deposits interbedded with subaqueous pyroclastic debris flow deposits (Cas and Wright, 1991). The presence of primary and reworked pyroclastic debris indicates penecontemporaneous volcanism and subsidence. A radiometric $^{40}Ar/^{39}Ar$ age of 16.5±0.5 Ma (F. Munizaga, written communication, 1991) obtained on plagioclase from a tuff near the top of the upper member constrains its minimum age to Early Miocene.

The Isla Manzano section (Fig. 3) has a thickness of 500 m, much greater than previously estimated. It consists of well-bedded tuff, tuffaceous and lithic sandstone, and tuffaceous siltstone deposited mainly by turbidity currents. Radiometric age determinations on samples from this locality yield no reliable results.

No chronological data are available for the Isla Puduhuapi strata. However, the presence of turbidite sequences similar to those from Isla Manzano suggests correlation of the two sections.

Several features of the Ayacara Formation are similar to those described for strike-slip basins (Christie-Blick and Biddle, 1988). These include:

(1) Geological mismatches with the surrounding area. For example, there appears to be no nearby source for clasts of andesite, dacite and shale, which reach diameters as great as 1 m in diameter in the lower member. and

(2) Rapid vertical facies changes. In its type locality, the upper member has deep water facies interfingered with strata containing algae, suggesting rapid subsidence of the basin (Fuenzalida, 1979).

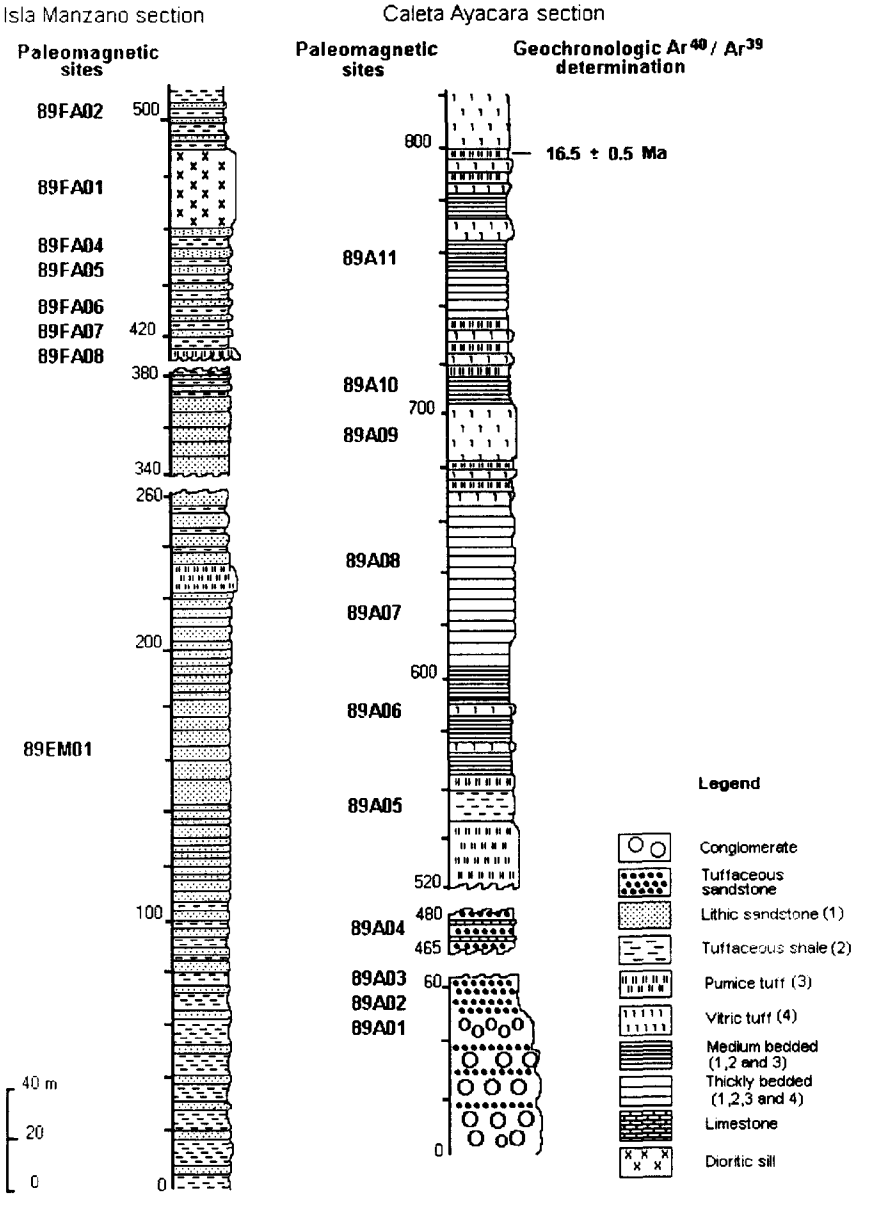


Figure 3. Stratigraphic sections of the Ayacara formation from Caleta Ayacara and Isla Manzano. Sampling sites for paleomagnetic and radiometric studies are indicated.

PALEOMAGNETIC STUDY

During regional geologic mapping of the Andean forearc, 24 paleomagnetic sites (165 samples) were obtained from three localities within the Ayacara Formation: 8 sites from Isla Manzano, 11 from Caleta Ayacara, and 5 from Isla Puduhuapi (Fig. 1). Six or more independently oriented samples were collected at each site. Samples were oriented using a magnetic compass, checked by solar compass when weather permitted. The very weak magnetizations of these rocks, and the fact that sun compass and magnetic compass orientations were in good agreement when both were available, indicate that local magnetic anomalies were not a problem.

In the laboratory magnetizations were measured on a Schonstedt spinner magnetometer. For each site, two representative specimens were selected to serve as pilot samples for preliminary thermal (T) and alternating field (AF) demagnetization. The AF demagnetization range used was 2.5 mT and 100 mT; thermal demagnetization was performed between 100° C and 680° C. All pilot demagnetizations included at least 10 steps. After pilot demagnetization, the remaining samples from each site were demagnetized by whichever method appeared to best define the characteristic remanent direction. Stable directions were identified visually as straight line segments of demagnetization paths on orthogonal vector diagrams and then determined analytically by principal component analysis (Kirschvink, 1980). Mean directions were calculated using the method of Fisher (1953), and also using the Bingham distribution (Onstott, 1980). As no significant difference between the two methods was detected, Fisher (1953) statistics are used throughout the study. Site-mean directions were converted to virtual geomagnetic poles (VGP), and final analysis was conducted using VGP.

To detect translations and/or rotations of sampling localities with respect to the stable interior of the continent, we must compare our paleomagnetic poles with a reference pole from the South American craton. This is surprisingly difficult to do, because there are very few well-determined poles of the correct age available on the South American craton. For instance, the compilation of Irving and Irving (1982) lists no South American reference poles at all between 64 and 15 Ma. For this reason we used the North American 20-40 Ma pole of Diehl *et al.* (1988) (81.5N, 147.3E, A-95 = 2.4°), rotating it into the South American reference framework using the 37 Ma fixed-hotspot stage pole of Engebretson *et al.*, (1985). Because our reference pole is for rocks that are predominantly younger than the youngest stage pole given by Engebretson *et al*, (1985) we pro-rated the rotation angle (using 2.3° instead of 4.3°). This gives a southern-hemisphere mid-Tertiary (20-40 Ma) reference pole of 82.0S, 308.3E (A-95 = 2.4°), only slightly different from the present spin-axis. Finally, rotations and poleward translations were calculated in pole-space following Beck (1989). Site-mean directions and statistics are summarized in Table 1; VGP calculations and statistics are given in Table 2.

Paleomagnetic Results

Here we first describe details of data selection, then discuss results by locality.

Data selection. Less than half the samples originally collected were used in the final averages. This resulted from acceptance criteria we applied at several levels. We rejected individual sample measurements if the estimate of precision during measurement was consistently low during repeat measurement (α-95 > 30°; Briden and Arthur, 1981). We also rejected some samples because their initial magnetic moments were too low (typically < 5 x 10 exp -5 A/m) for us to obtain reliable demagnetizations paths in reasonable time. We also rejected individual sample measurements if the precision of line-fit was low (MAD > 20°; Kirschvink, 1980). Entire sites were rejected if their circles of 95% confidence (α-95) exceeded 20°. In calculating site-mean directions we used anchored rather than free lines. Comparison of the two indicated that no systematic high-temperature/high-coercivity component was present (no systematic difference in mean direction was detected), and means calculated using anchored lines tended to be more precise. After application of these selection criteria we were left with 11 of our original 24 sites, representing 72 of the original 165 samples. Thus, the attrition rate on this project was high, but internal (within-site) and external (between-site) consistency indicates that we have measured a meaningful remanent direction.

Isla Manzano

Four (of 8) sites from this locality passed the screening criteria; these consisted of three sedimentary horizons and a sill. Stable directions in the sedimentary layers were isolated by thermal demagnetization, usually between 415° and 515° C (Fig. 4a). AF demagnetization was less effective in removing a low-stability component, which tended to be northward and down (Fig. 4a,b). In the sill a single stable component of magnetization was isolated by AF demagnetization between 20 mT and 80 mT and also by thermal demagnetization between 100° and 580° C. Site-mean directions from the sill and sedimentary rocks are similar; very shallow and northwesterly before tilt-correction, much steeper and northwesterly after tilt-correction. Correction for tilt causes a slight increase in dispersion (k, the precision parameter of Fisher (1953) decreases from 200 to 120). All sites have normal magnetic polarity.

Caleta Ayacara

Eleven sites were obtained from this locality but only five gave reasonably reliable results, and two of these were finally rejected because their circles of confidence slightly exceeded 20° (Table 1). Most samples had a low coercivity component that was removed by AF demagnetization at peak fields of up to 30 mT (Fig. 4c). At higher fields (generally 30 to 80 mT) a steady trend toward the origin indicated isolation of a single stable component. Thermal demagnetization between 415° C to 580° C gave a similar path, suggesting that titanomagnetite and/or magnetite car-

Table 1. Site-mean directions of remanent magnetization for the Ayacara Fm., southern Chile

Site	D//lu	D/Ic	N/No	R	α-95	Az/Dip
Isla Manzano (42.02S, 72.67W)						
89EM01	325.4/-32.7	28.7/-79.3	7/7	6.913	7.3	313/54
89FA01	318.7/-24.4	320.1/-72.4	9/9	8.758	9.1	318/48
89FA05	328.1/-32.7	357.2/-78.7	7/7	6.963	4.7	320/48
89FA06	329.0/-27.4	349.6/-73.7	6/6	5.904	9.4	320/48
Mean-u	325.2/-29.4		4/4	3.985	6.6	
Mean-c		349.2/-77.2	4/4	3.975	8.4	
Caleta Ayacara (42.30S, 72.80W)						
89A01*	74.5/8.9	87.5/50.2	4/8	3.862	20.1	233/46
89A04	98.2/16.4	138.4/56.8	6/6	5.979	4.4	245/60
89A05*	281.5/-14.9	319.2/-54.9	5/5	4.685	22.3	247/60
89A06	292.0/-26.1	4.0/-50.2	6/6	5.973	5.0	251/75
89A09	89.3/17.2	169.3/59.1	7/7	6.979	3.6	239/82
Mean-u	279.6/-20.1		3/5	2.957	18.3	
Mean-c		345.1/-56.9	3/5	2.941	21.5	
Isla Puduhuapi (42.93S, 72.79W)						
89FU58	187.1/51.7	168.1/43.4	5/5	4.964	7.3	113/19
89FU59	10.4/-63.2	3.0/-54.2	6/6	5.931	7.9	160/10
89FU60	150.9/67.6	153.5/57.7	6/6	5.962	5.9	160/10
90PA28	166.6/66.2	144.8/51.9	7/7	6.943	5.9	113/19
Mean-u	356.1/-63.0		4/4	3.945	12.5	
Mean-c		347.9/-55.8	4/4	3.954	11.5	
Cocotue Beach Basalts (41.92S, 74.02W)						
85CJ35		168.2/70.7	6/6	5.985	3.7	
85CJ36		143.6/64.6	6/6	5.995	2.1	
85CJ37		137.7/52.0	6/6	5.904	9.4	
Mean		326.8/-63.0	3/3	2.958	18.1	

D/lu, mean declination and inclination of site, uncorrected for tilt; D/Ic, same, corrected for tilt; N/No, number of samples used on calculation/number originally collected; R, length of vector sum of N unit vectors; α-95, radius of circle of 95% confidence. Sites marked with *not used in mean calculations.

ries the magnetization (Fig. 4d). Two sites (89A05, 06) had upward (normal polarity) directions; these lie stratigraphically between other sites that have downward (reverse polarity) directions. Thus we apparently sampled through at least two geomagnetic reversals. Scatter increases very slightly when the tilt correction is made (k decreases from 46.5 to 33.9).

Isla Puduhuapi

Four of five sites from this location gave reliable directions. Sites 89FU58, 59, and 60 show similar AF demagnetization paths. A low coercivity component is removed at peak fields of 20 mT, and a single component is described thereafter, to peak fields of 70 mT. Site 90PA28 shows a high-coercivity component (probably carried by pyrrhotite) which could not be removed by either AF (Fig. 4e) or thermal demagnetization alone. However, a combination of thermal demagnetization to 350° C, followed by AF demagnetization to 70 mT, effectively isolated a stable direction (Fig. 4f). This locality consists of three sedimentary layers and one dike; sedimentary layers have reverse polarity but the dike has normal polarity. Correcting for tilt (including a correction for the dike based on the attitude of its enclosing sedimentary layers) slightly decreases scatter (k increases from 54.5 to 65.2). We also computed a mean direction under the assumption that the dike was intruded after the sedimentary layers were tilted, but this result (D, 342.8; I, -55.1; k, 35.0) is more scattered than either the all-corrected or all-in situ scenarios.

Table 2. Site-mean VGP and statistics for the Ayacara Fm, southern Chile.

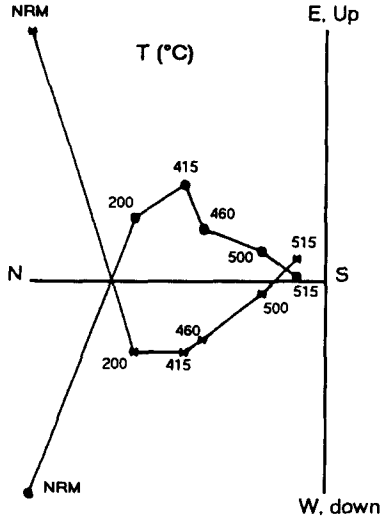
Site	Lat/Long-u	Lat/Long-c	N	R	A-95
		Isla Manzano			
89EM01	51.7/46.6	59.4/267.7			
89FA01	43.6/44.6	60.9/332.1			
89FA05	53.4/49.7	64.2/289.6			
89FA06	51.4/54.2	71.7/304.1			
Mean-u	50.1/48.6		4	3.988	5.9
Mean-c		66.1/297.5	4	3.919	15.3
		Caleta Ayacara			
89A01*	8.3/3.4	18.6/351.9			
89A04	11.7/16.5	58.0/11.9			
89A05*	13.6/19.3	57.8/15.8			
89A06	25.4/21.6	78.2/124.2			
89A09	5.4/10.2	81.6/30.5			
Mean-u	14.2/15.9		3	2.959	17.8
Mean-c		77.3/33.7	3	2.887	30.2
		Isla Puduhuapi			
89FU58	78.0/137.3	76.3/96.1			
89FU59	82.3/213.8	81.4/124.0			
89FU60	68.8/346.2	69.5/21.2			
90PA28	79.1/341.5	70.3/14.2			
Mean-u	85.9/343.3		4	3.893	17.6
Mean-c		78.4/46.2	4	3.918	15.4
		Cocotue Beach Basalts			
85CJ35		74.7/312.5			
85CJ36		63.7/353.1			
89CJ37		55.5/17.4			
Mean			3	2.926	24.3

Lat/Long-u, south latitude/east longitude of site VGP, calculated from entries in Table 1, uncorrected for tilt; Lat/Long-c, same, corrected for tilt; N, R as in Table 1. A95, radius of 95% confidence about mean pole.

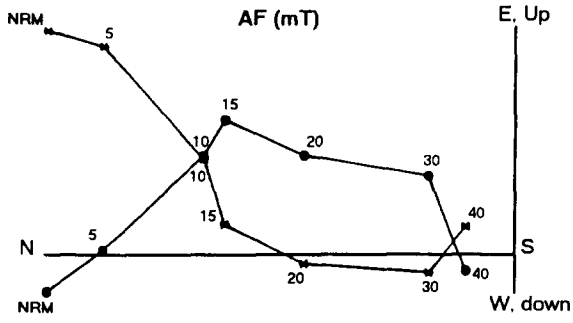
DISCUSSION

Isla Manzano

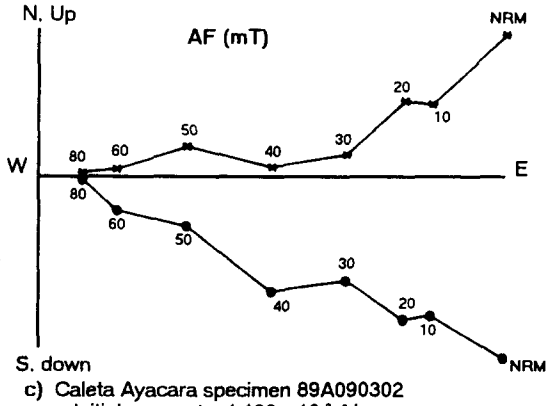
The four sites from Isla Manzano all have normal magnetic polarity and between-site dispersion is so low that it appears unlikely that the geomagnetic secular variation has been averaged. Dispersion increases when the tilt-correction is made, which might suggest that the magnetization was acquired after tilting; that is, that the rocks are remagnetized. However, the change in dispersion is not significant at 95% confidence, using the chi-squared test of Cox (1969). Furthermore, there are no obvious signs of reheating or alteration, such as might suggest remagnetization. We suspect that the increase in scatter when the tilt-correction is made is simply the result of inaccuracy in measuring true strike and dip. Given the rotated Diehl *et al* (1983) reference pole cited above, the direction one would expect to observe for mid-Tertiary rocks on Isla Manzano (assuming no post-magnetization displacement with respect to the South American craton) is: $D_x = 355.5$; $I_x = -67.0$. (Here the subscript x denotes the "expected direction"; Beck, 1980). Both in situ and tilt-corrected mean directions for Isla Manzano have declinations that are rotated counterclockwise with respect to the expected direction. However, the in situ direction has an inclination that is extremely shallow, such as would be observed (for instance) if the rocks had moved several thousand kilometers southward relative to the South American craton. However, if the tilt correction is made, observed and expected inclinations are very similar and no large coastwise displacements are required. It is very unlikely on



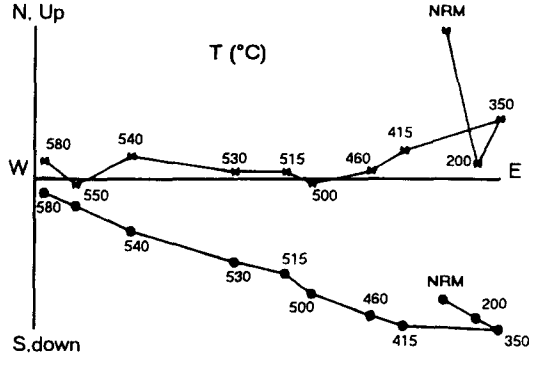
a) Isla Manzano specimen 89EM010701
Initial moment = 3.307 x 10^{-4} A/m



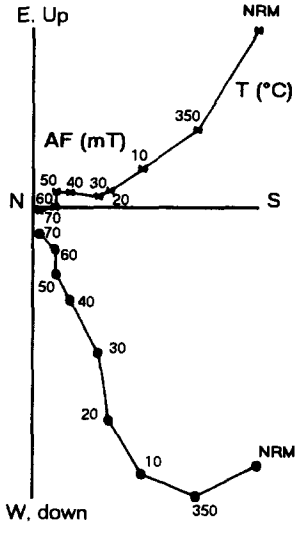
b) Isla Manzano specimen 89EM010702
Initial moment = 5.401 x 10^{-4} A/m



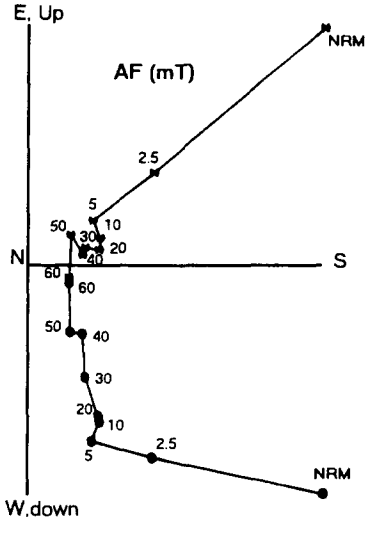
c) Caleta Ayacara specimen 89A090302
Initial moment = 1.130 x 10^{-3} A/m



d) Caleta Ayacara specimen 89A090301
Initial moment = 1.039 x 10^{-3} A/m



e) Isla Puduhuapi specimen 90PA2807
Initial moment = 3.608 x 10^{-3} A/m



f) Isla Puduhuapi specimen 90PA2806
Initial moment = 3.826 x 10^{-3} A/m

Figure 4. Typical demagnetization diagrams from samples of the Ayacara Formation. (a and b) Isla Manzano; (c and d) Caleta Ayacara; (e and f) Isla Puduhuapi. Not corrected for tilt. x= map view; o= vertical section.

several grounds that Isla Manzano has been displaced several thousand kilometers southward since the mid-Tertiary, hence we will use the tilt-corrected directions for this locality.

Caleta Ayacara

As mentioned earlier, the polarity of the Caleta Ayacara sites defines an R-N-R sequence. Thus we have sampled across two polarity transitions, which suggests that our sites are spread out enough in time to adequately represent the longer frequencies of the geomagnetic secular variation. Dispersion increases when the tilt correction is made, but the increase is not significant. Expected direction for this locality is: D_x = 355.6; I_x = -67.0. Again, both in situ and tilt-corrected mean declinations are rotated counterclockwise with respect to the reference direction. Both inclinations are more shallow than expected, but the in situ mean inclination is improbably shallow. As there is no geologic evidence to suggest remagnetization we will use the tilt-corrected data set in future calculations.

Isla Puduhuapi

The Isla Puduhuapi data set consists of three sedimentary units and a dike. The dike clearly is of a different age than the sedimentary layers because it has a different polarity. However, the dike's direction of remanent magnetization falls within the grouping defined by the sedimentary units and there seems to be no reason to omit it from the final averaging process. Dispersion decreases slightly when the tilt-correction is applied, but the change is not statistically significant. The expected direction for this locality is: D_x = 355.5; I_x = -67.5, very nearly identical to the mean in situ direction (Table 1). The tilt-corrected mean has a slightly shallow inclination and a declination that is rotated counterclockwise. Based on the slight decrease in dispersion when the tilt-correction is applied, the existence of two polarities, and the lack of any geological evidence for alteration or reheating, we use the tilt-corrected directions in further calculations.

To calculate a paleomagnetic pole for the Ayacara Formation we averaged the 11 VGP listed in Table 2. This gives a mean pole at 79.6S, 344.9E, with a circle of 95% confidence (A-95) of 11.5° and a precision parameter (k) of 16.9. A similar mean using poles from in situ directions (54.8S, 29.1E) has much greater dispersion (K = 6.2) and a highly elongate distribution. The difference in precision is significant at 99% confidence (Cox, 1969). Overall precision is somewhat low, probably reflecting random errors in measuring attitudes at the outcrop.

Tables 1 and 2 also contain paleomagnetic results for three sites of Cocotue Beach basalts (Fig. 1). These results (early Miocene: 21.7±1.7 Ma; García *et al.*, 1988) are the only other paleomagnetic data available from west of the LOFZ. The expected direction at Cocotue Beach, based on the rotated Diehl *et al.*, (1988) reference pole, is: D_x = 355.4; I_x = -66.6. The observed direction (Table 1) thus is concordant in inclination but rotated sharply counterclockwise in declination.

By including the Cocotue Beach data we can calculate a mean VGP for middle Tertiary rocks west of the LOFZ. This lies at 75.8S, 353.6E (A-95 = 10.0°; K = 16.4), Fig. 5b.

In Fig. 5a we show virtual geomagnetic poles for the four localities discussed in this section. Locality mean VGP are shown in Fig. 5b, together with the overall mean pole and the rotated Diehl *et al.* (1988) reference pole. Clearly, all locality VGP and the resultant overall pole lie to the left of the reference pole, as viewed from the sampling area. This suggests that the region west of the Liquiñe-Ofqui fault has rotated counterclockwise relative to stable South America. The amount of rotation can be calculated from the reference pole and the calculated mean VGP using the method of Beck (1989). For a location near the common center of the four sampling localities (42.25S, 287E) the rotation is 14.6°±11.9° counterclockwise. Poleward transport is negligible; 3.3°±8.1° (error limits calculated after Demarest, 1983). These error limits clearly are minimum values, because any error that might arise from rotating the Diehl *et al* (1988) North American reference pole into South American coordinates is not included.

TECTONIC INTERPRETATION

The counterclockwise rotation shown by the Ayacara and Cocotue Beach rocks is small, but it appears to be real and tectonically significant. Westerly declinations (indicating counterclockwise rotations) are found throughout a broad area west of the LOFZ, whereas declinations measured east of the LOFZ lie consistently clockwise of the expected direction (Cembrano *et al*, 1992). Moreover, most rocks spatially associated with the LOFZ in the Lake District, several hundred kilometers to the north, also are rotated clockwise (García *et al.*, 1988). Thus tectonic processes operating west of the LOFZ in our field area are different in an important way from processes normally associated with dextral shear.

In another paper we propose and describe in detail a speculative model that can account for counterclockwise rotations outboard of a dextral shear zone at the southern end of the LOFZ, as well as for other instances of "wrong-way" rotation in zones of strike-slip faulting (Beck *et al.*, 1993). Here we will outline the model as it applies to the southern LOFZ. The model is suggested by the striking pattern of faults and other lineaments found along the southern end of the LOFZ. We believe that this instance of wrong-way rotation arises because the crustal sliver outboard (west) of the fault is prevented from freely moving northward by the presence of a buttress (Beck, 1991).

The model is explained in Fig. 6. North-oblique subduction creates the conditions necessary to detach an outboard sliver from the leading edge of South America. However, owing perhaps to a steepening of subduction angle, or to a gradual decrease in the angle of obliquity, these conditions die out northward. A buttress therefore is present, preventing free movement of the sliver. The pattern of fjords and other lineaments in the southern LOFZ region suggests that northward displacement has been

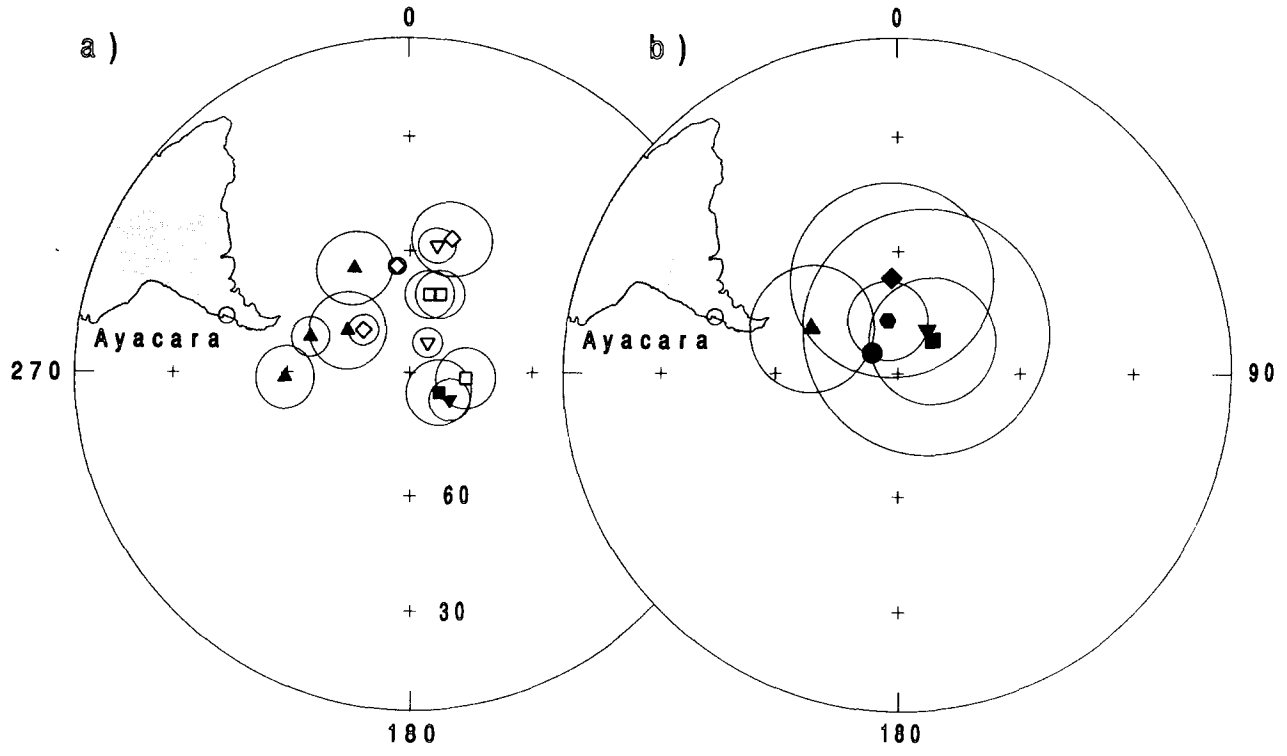


Figure 5. Equal area projections of paleomagnetic poles from the Ayacara Formation (location shown by circle). (a) Site-mean virtual geomagnetic poles from the Ayacara Formation (VGPs) coded by locality: Isla Manzano (triangle), Caleta Ayacara (inverted triangle), Isla Puduhuapi (square), and Cocotue Beach basalts (diamond). Solid symbols from normal polarity magnetizations; open symbols, reverse. (b) Locality VGPs, overall mean pole (hexagon) and 20-40 Ma reference pole (filled circle). Elipses around all poles show 95% confidence limits (A-95; Fisher, 1953).

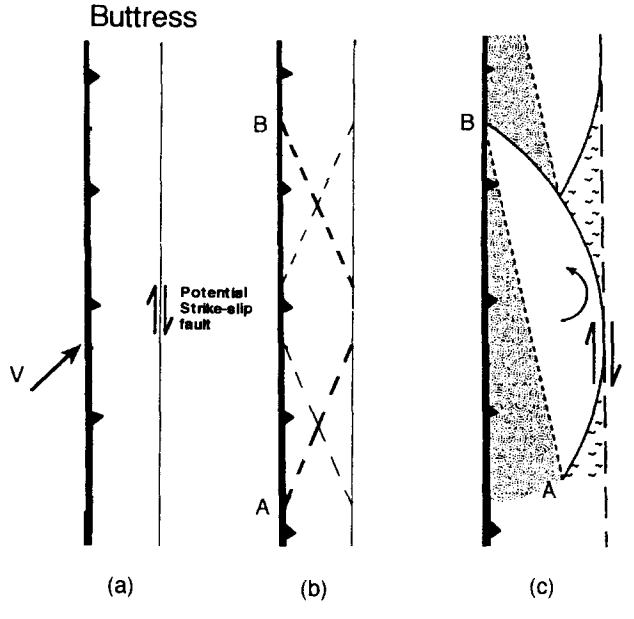


Figure 6. Overcoming a buttress by widening the sliver (Beck *et al.* 1993) (a) sliver is impelled northward by oblique subduction but it is prevented from moving by a buttress. Trench is outlined by a thick, barbed line. Incipient master strike-slip fault shown by thin line. (b) resulting shortening produces conjugate fractures (dashed lines). Lens-shaped blocks form taking advantage of pre-existing shears. (c) Lens-shaped blocks move northward by rotating counterclockwise and overlapping one another; sliver widens thereby. Areas of potential sedimentary basins are shown by dotted pattern, wiggle pattern shows areas of shortening and potential uplift.

accommodated by segmenting the sliver along crescent shaped faults, then sliding the independent block so created northward, rotating them counterclockwise in the process. The map pattern so created suggests imbrication, but in the horizontal plane and at a colossal scale. Many other examples of curvature in strike-slip fault systems are known (e.g., the Atacama fault in northern Chile and the faults of southeastern Alaska), and we wonder if this mechanism may have application elsewhere.

A CAUTIONARY NOTE

The results presented in this article must be considered preliminary, and the model we present thus is highly speculative. In Fig. 5a the VGP representing normal-polarity and reverse-polarity sites are distinguished and it is clear that there is a difference between the two polarity-subsets. This could simply be an accident of locality, but it could also indicate that, despite careful laboratory demagnetization, some unremoved secondary magnetic component still remains. In the latter case the evidence for counterclockwise rotation of crustal blocks west of the LOFZ is weakened, and our model severely compromised. However, by presenting our data and model at this preliminary stage, we hope to encourage further paleomagnetic research on rock units having similar tectonic setting along the LOFZ or on other major strike-slip fault systems worldwide.

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